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Quantifying non-Markovianity: a quantum resource-theoretic approach

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The quantification and characterization of non-Markovian dynamics in quantum systems is an essential endeavor both for the theory of open quantum systems and for a deeper understanding of the effects of non-Markovian noise on quantum technologies. Here, we introduce the robustness of non-Markovianity, an operationally-motivated, *optimization-free* measure that quantifies the minimum amount of Markovian noise that can be mixed with a non-Markovian evolution before it becomes Markovian. We show that this quantity is a bonafide non-Markovianity measure, since it is faithful, convex, and monotonic under composition with Markovian channels. A two-fold operational interpretation of this measure is provided, with the robustness measure quantifying an advantage in both a state discrimination and a channel discrimination task. Furthermore, we provide a closed-form analytical expression for this measure and show that, quite remarkably, the robustness measure is exactly equal to half the Rivas-Huelga-Plenio (RHP) measure [Phys. Rev. Lett. **105**, 050403 (2010)]. As a result, we provide a direct operational meaning to the RHP measure while endowing the robustness measure with the physical characterizations of the RHP measure.

I. INTRODUCTION

The idealization of a quantum system coupled to a *memoryless* environment is exactly that: an idealization. In the theory of open quantum systems, memoryless or Markovian evolution arises from the assumptions of weak (or singular) coupling to a fast bath [1, 2]. Although very useful, this assumption does not always apply to physical systems of interest—including, but not limited to, several quantum information processing technologies—and so a complete physical picture cannot neglect non-Markovian effects. Markov processes, although well-defined classically, have several distinct definitions and characterizations in the quantum case, leading to different definitions of non-Markovianity (see Refs. [3, 4] for a detailed comparison). Prominent examples include the semigroup formulation [5], the distinguishability measure [6], and CP-divisibility [7]. In this paper, we use the CP-divisibility approach to non-Markovianity.

More formally, the time evolution of a quantum dynamical system is called Markovian (or divisible [8]) if there exists a family of trace-preserving linear maps, $\{\mathcal{E}_{(t_2,t_1)}, t_2 \geq t_1 \geq t_0\}$ which satisfy the composition law, $\mathcal{E}_{(t_3,t_1)} = \mathcal{E}_{(t_3,t_2)} \circ \mathcal{E}_{(t_2,t_1)}, \quad t_3 \geqslant t_2 \geqslant t_1$, where $\mathcal{E}_{(t_2,t_1)}$ is a CP map for every t_2 and t_1. Quantum systems undergoing Markovian dynamics are described by a master equation, as described by the following fundamental result (under the assumptions of a *differentiable* quantum Markovian process, see Ref. [3] for more details):

Theorem I.1 (**Gorini-Kossakowski-Sudarshan-Lindblad** [9–11])**.** *An operator $\mathcal{L}_t$ is the generator of a quantum Markov (or divisible) process if and only if it can be written in the form*

$$\frac{\mathrm{d}\rho(t)}{\mathrm{d}t} = \mathcal{L}_t[\rho(t)] = -\mathrm{i}[H(t), \rho(t)] + \sum_k \gamma_k(t) \left[V_k(t)\rho(t)V_k^\dagger(t) - \frac{1}{2}\left\{ V_k^\dagger(t)V_k(t), \rho(t) \right\} \right], \tag{1}$$

where $H(t)$ and $V_k(t)$ are time-dependent operators, with $H(t)$ self-adjoint, and $\gamma_k(t) \geq 0$ for every k and t.

Recently, there has been some effort to characterize, both qualitatively and quantitatively, the *resourcefulness* of non-Markovianity in quantum information processing tasks [12], quantum thermodynamics [13], and quantum error-suppression techniques like dynamical decoupling [14]. However, quantum information theory already has a powerful formalism to characterize the quantification and manipulation of a quantum resource, called *quantum resource theories* [15]. In this paper, looking through the lens of quantum resource theories, we analyze quantum non-Markovianity as a resource.

The construction of operational measures for quantum resources is one of the many goals of quantum resource theories. In this direction, there has been a lot of exciting work recently, especially with regard to what are called *robustness* measures, which characterize how robust a resource is with respect to "mixing." Robustness measures with operational significance have been constructed for quantum entanglement [16], coherence [17–19], asymmetry [20], and other resources. Notably, it was recently shown that for any quantum resource that forms a convex resource theory, there exists a subchannel discrimination game where that resource provides an advantage that is quantified by a generalized robustness measure

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[21]. These results were further generalized to arbitrary convex resource theories—both in quantum mechanics and general probabilistic theories—characterizing the resource content not only of quantum states but also of quantum measurements and channels [22] (see also related work by Ref. [23]).

Motivated by the operational nature of robustness measures in quantum resource theories, we construct a robustness measure for non-Markovianity, which quantifies the minimum amount of Markovian "noise" that needs to be mixed (in the sense of convex combination) with a non-Markovian process to make it Markovian. A few remarks are in order before we go into more detail, especially about the nature of this resource. Quantum non-Markovianity is a *dynamic* resource, as opposed to *static* resources like coherence, entanglement, magic, etc. Static resources are a property of quantum *states* while dynamic resources are a property of the quantum *process*. We believe that a proper characterization of a dynamic resource like non-Markovianity needs to identify the equivalent of *free operations* in theories of static resources [24]. We define the set of Markovian channels as the free operations, and we characterize the resourcefulness of non-Markovian channels with respect to this set. In this resource-theoretic construction for non-Markovianity, there are no *free states* per se, in contrast to the typical approach in static quantum resource theories. In Refs. [25, 26], the authors construct a resource theory for non-Markovianity where they identify the set of free states as the Choi-Jamałkowski matrices corresponding to Markovian channels. Although this construction is in some ways easier to work with, since there are both designated free states and free operations, this quickly leads into difficulties. Any resourceful operation can transform a free state (a Choi matrix for a Markovian channel, which is normalized and positive semidefinite) into a non-state (since it can now have negative eigenvalues) and so it can take us outside the set of quantum states, which is usually assumed to be the landscape for quantum resource theories. To circumvent such obstacles, we avoid such an identification and only define a set of free operations.

This paper is organized as follows. In Section II, we briefly review some details about non-Markovianity and a proof that the set of Choi matrices corresponding to Markovian channels is closed and convex in the short-time limit. In Section III, we introduce the robustness of non-Markovianity and show that this quantity is in fact, a *bonafide* measure in that it is faithful, convex, and monotonic under composition with Markovian channels. We also cast the problem of computing this measure as a semidefinite program, which will be useful later. In Section IV, we provide a two-fold operational meaning to this measure by explicitly constructing both state and channel discrimination tasks where the advantage is proportional to the robustness measure. In Section V, we relate the robustness measure to the "gold standard," for the quantification of non-Markovianity (in the CP-divisibility framework), the RHP measure, and show quite remarkably that it is equal to half the RHP measure. We study some examples in Section VI and conclude with final remarks in Section VII.

II. PRELIMINARIES

A. The Choi-Jamiołkowski isomorphism

Let $\mathcal{H}$ be a finite-dimensional Hilbert space. Then, consider the (normalized) maximally entangled state between two copies of the Hilbert space, $|\Phi^+\rangle = \frac{1}{\sqrt{d}}\sum_{i=0}^{d}|i\rangle|i\rangle$, where d is the dimension and $\{|i\rangle\}_{i=0}^{d-1}$ an orthonormal basis for the Hilbert space. To a linear map $\mathcal{E}:\mathcal{H}\to\mathcal{H}$, we associate a Choi-Jamiołkowski matrix (called a Choi matrix for brevity), defined as

$$\rho_{\mathcal{E}} \equiv [\mathbb{I}\otimes\mathcal{E}]\left(|\Phi^+\rangle\langle\Phi^+|\right). \tag{2}$$

Then, $\rho_{\mathcal{E}}$ is positive-semidefinite if and only if the map $\mathcal{E}$ is completely positive [27, 28]. And, $\|\rho_{\mathcal{E}}\|_1 = 1$ if $\mathcal{E}$ is trace-preserving. See the Appendix A for more details about the Choi-Jamiołkowski isomorphism.

B. The short-time limit of Lindblad dynamics

In this paper we only consider dynamics with Lindblad-type generators (see Eq. (1)), primarily for finite-dimensional systems, although several results generalize to the infinite-dimensional case. In Ref. [25], it was shown that the set of all Markovian Choi matrices, i.e., the Choi matrices corresponding to Markovian channels, form a convex and compact set in the short-time limit.

Theorem II.1 ([25]). *The set of all Markovian Choi matrices,* $\mathcal{F}^{\epsilon,t} = \left\{\rho_{\mathcal{E}}(t+\epsilon,t) \mid \left\|\rho_{\mathcal{E}}(t+\epsilon,t)\right\|_1 = 1, \forall t, \epsilon > 0\right\}$ *is a convex and compact set in the limit* $\epsilon\to 0^+$.

Proof. See the Appendix B for a proof of this theorem. ■

In light of the above theorem, we define the set of all Markovian channels (at some time t) in the limit $\epsilon\to 0^+$ as the *free operations*. Using the Choi-Jamiołkowski isomorphism, this makes $\mathcal{F}^{\epsilon,t}$ the set of *free operations* when we are working in the Choi matrix representation. Intuitively, any physically-motivated distance measure (that is say, contractive under CPTP maps) from this set would suffice as a quantifier of non-Markovianity.

III. ROBUSTNESS OF NON-MARKOVIANITY

We now introduce the robustness of non-Markovianity measure. Define $\mathcal{O}_{\mathcal{F}}(t,\epsilon)\equiv\mathcal{O}_{\mathcal{F}}$ as the set of free channels, which (courtesy of the theorem above) is closed and convex. We drop the t,ϵ below for brevity. We define the *robustness of*

non-Markovianity (RoNM) as

$$\mathcal{N}_{\mathcal{R}}(\Lambda) = \min_{\Phi \in \mathcal{O}_{\mathcal{F}}} \left\{ s \geq 0 \mid \frac{\Lambda + s\Phi}{1+s} =: \Gamma \in \mathcal{O}_{\mathcal{F}} \right\}. \quad (3)$$

We emphasize that since the "mixing" is with respect to $\mathcal{O}_{\mathcal{F}}$, i.e., $\Phi \in \mathcal{O}_{\mathcal{F}}$, this is not the *generalized* robustness measure introduced in Ref. [22]. This choice will play a crucial role in the operational interpretation of this measure, as we'll see in Section IV. An equivalent form for the RoNM can be obtained using the Choi-Jamiołkowski representation. Given the Choi matrix ρ_Λ corresponding to a map Λ and setting $\rho \equiv \rho_\Lambda$, its RoNM is obtained as

$$\mathcal{N}_{\mathcal{R}}^{\text{Choi}}(\rho) = \min_{\tau \in \mathcal{F}} \left\{ s \geq 0 \mid \frac{\rho + s\tau}{1+s} =: \delta \in \mathcal{F} \right\}, \quad (4)$$

where $\mathcal{F} \equiv \mathcal{F}^{t,\epsilon}$ is the set of all Markovian Choi matrices. (Once again, we drop the superscripts for brevity.) For a fixed channel Λ, $\mathcal{N}_{\mathcal{R}}(\Lambda) = \mathcal{N}_{\mathcal{R}}^{\text{Choi}}(\rho_\Lambda)$. Therefore, we don't make the distinction unless it is necessary, and define $\mathcal{N}_{\mathcal{R}}^{\text{Choi}}(\cdot) = \mathcal{N}_{\mathcal{R}}(\cdot)$.

From the definition of the RoNM, one can define an *optimal decomposition* of a channel as

$$\Lambda = (1+s^\star)\,\Gamma^\star - s^\star \Phi^\star, \quad (5)$$

where $s^\star = \mathcal{N}_{\mathcal{R}}(\Lambda)$ and $\Gamma^\star, \Phi^\star \in \mathcal{O}_{\mathcal{F}}$. Similarly, in the Choi matrix representation, we have an optimal decomposition:

$$\rho = (1+s^\star)\,\delta^\star - s^\star \tau^\star, \quad (6)$$

where $s^\star = \mathcal{N}_{\mathcal{R}}(\rho)$ and $\delta^\star, \tau^\star \in \mathcal{F}$.

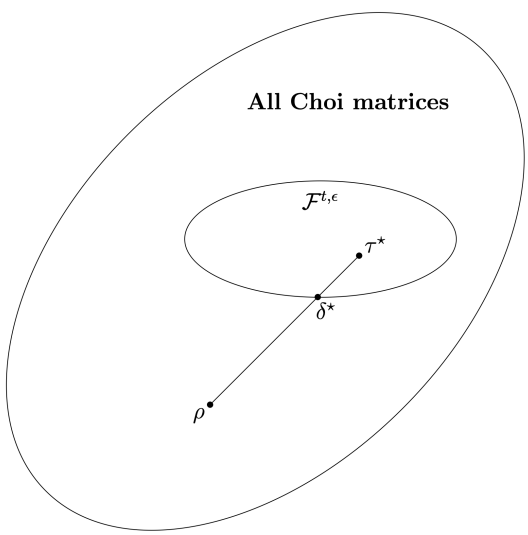

Figure 1. Robustness of non-Markovianity for a Choi matrix ρ. The outer ellipse is the set of all Choi matrices and $\mathcal{F}^{t,\epsilon}$ is the set of Markovian Choi matrices. The $\tau^\star, \delta^\star$ correspond to the optimal decomposition.

We now show that this measure has the properties of faithfulness, convexity, and monotonicity.

Faithfulness.—If $\Lambda \in \mathcal{O}_{\mathcal{F}}$, then clearly $\mathcal{N}_{\mathcal{R}}(\Lambda) = 0$, since we do not need to mix any amount of Markovian noise to make it Markovian. Conversely, $\mathcal{N}_{\mathcal{R}}(\Lambda) = 0 \implies \Lambda \in \mathcal{O}_{\mathcal{F}}$.

Convexity.—Given two Choi matrices ρ_1 and ρ_2 with optimal decompositions (see Eq. (6)), we have $\rho_k = \left(1 + \mathcal{N}_{\mathcal{R}}(\rho_k)\right)\delta_k^\star - \mathcal{N}_{\mathcal{R}}(\rho_k)\tau_k^\star$, where $k = 1, 2$. Then, a convex combination of these two matrices, $\rho = p\rho_1 + (1-p)\,\rho_2$, with $0 \leq p \leq 1$ can be written as $\rho = (1+s)\,\delta - s\tau$, by choosing

$$\delta = \frac{p\left(1+\mathcal{N}_{\mathcal{R}}(\rho_1)\right)\delta_1^\star + (1-p)\left(1+\mathcal{N}_{\mathcal{R}}(\rho_2)\right)\delta_2^\star}{(1+s)} \in \mathcal{F},$$
$$\tau = \frac{p\mathcal{N}_{\mathcal{R}}(\rho_1)\tau_1^\star + (1-p)\,\mathcal{N}_{\mathcal{R}}(\rho_2)\tau_2^\star}{s} \in \mathcal{F}, \text{ and}$$
$$s = p\mathcal{N}_{\mathcal{R}}(\rho_1) + (1-p)\,\mathcal{N}_{\mathcal{R}}(\rho_2).$$

And since $\mathcal{N}_{\mathcal{R}}(\rho) \leq s$, we have

$$\mathcal{N}_{\mathcal{R}}\left(p\rho_1 + (1-p)\,\rho_2\right) \leq p\mathcal{N}_{\mathcal{R}}(\rho_1) + (1-p)\,\mathcal{N}_{\mathcal{R}}(\rho_2). \quad (7)$$

A similar proof follows for the robustness of the channel corresponding to the Choi matrix ρ_Λ.

Monotonicity under free operations.—The free operations in this setting are the Markovian maps, which are *quantum channels* or *completely positive trace-preserving* (CPTP) maps. Since there are no free states in this construction, we instead analyze the monotonicity under composition with free operations. Consider the composition of a non-Markovian map, say, Λ, with a free operation, say Γ, i.e., $\Gamma \circ \Lambda \equiv \Gamma(\Lambda)$. Then, using the closed form expression for $\mathcal{N}_{\mathcal{R}}(\Phi)$ in terms of its corresponding Choi matrix ρ_Φ (which we derive later, see Eq. (23)), we have,

$$\begin{aligned}
\mathcal{N}_{\mathcal{R}}(\Gamma \circ \Lambda) &= \frac{\left\|\rho_{(\Gamma\circ\Lambda)}\right\|_1 - 1}{2}, \\
&= \frac{\left\|(\mathbb{I}\otimes\Gamma)\left\{\mathbb{I}\otimes\Lambda\left(|\Phi^+\rangle\langle\Phi^+|\right)\right\}\right\|_1 - 1}{2}, \\
&= \frac{\left\|(\mathbb{I}\otimes\Gamma)\,\rho_\Lambda\right\|_1 - 1}{2}, \\
&\leq \frac{\|\rho_\Lambda\|_1 - 1}{2} = \mathcal{N}_{\mathcal{R}}(\Lambda),
\end{aligned} \quad (8)$$

where in the last inequality, we've used the fact that if $\mathcal{E}$ is a quantum channel and $\Delta \in \mathcal{B}(\mathcal{H}\otimes\mathcal{H})$ is Hermitian, then, $\left\|(\mathbb{I}\otimes\mathcal{E})(\Delta)\right\|_1 \leq \|\Delta\|_1$ (see Ref. [3, 29] for more details). Therefore, the RoNM measure is monotonic under composition with free operations. Together, these imply that the RoNM is a faithful, convex, and monotonic measure of quantum non-Markovianity. We now cast the RoNM as a semidefinite program [30].

A. Semidefinite program for the RoNM

A semidefinite program (SDP) is a triple (Φ, A, B), where, Φ is a hermiticity-preserving map from the Hilbert space $\mathcal{X}$ to $\mathcal{Y}$ and A, B are Hermitian matrices over the Hilbert spaces $\mathcal{X}, \mathcal{Y}$, respectively [31]. Then, we can define a pair of optimization problems associated with the SDP, called the primal and dual problems

$$\begin{array}{cc} \underline{\text{Primal problem}} & \underline{\text{Dual problem}} \\ \begin{array}{c}\text{maximize: } \langle A, X\rangle, \\ \text{subject to: } \Phi(X) \leq B, \\ X \geq 0.\end{array} & \begin{array}{c}\text{minimize: } \langle B, Y\rangle, \\ \text{subject to: } \Phi^*(Y) \geq A, \\ Y \geq 0.\end{array} \end{array} \tag{9}$$

We now show that the robustness measure can be cast as a SDP. Given a Choi matrix ρ, a decomposition of the form $\rho = (1+s)\,\delta - s\tau$ is equivalent to $\rho \leq (1+s)\,\delta$, where $\delta \in \mathcal{F}$, since there exists a (Markovian) Choi matrix τ, such that, $\rho - (1+s)\,\delta = s\tau$ if $\rho \leq (1+s)\,\delta$. Then, the RoNM can be characterized as,

$$\mathcal{N}_{\mathcal{R}}(\rho) = \min_{\delta \in \mathcal{F}} \left\{ s \geq 0 \mid \rho \leq (1+s)\,\delta \right\}. \tag{10}$$

In light of this, we can define the semidefinite programming form of $\mathcal{N}_{\mathcal{R}}(\rho)$ as

$$\begin{aligned} \text{minimize:} \quad & \lambda - 1 \\ \text{subject to:} \quad & \rho \leq \delta, \\ & \delta \geq 0, \\ & \mathrm{Tr}\left[\delta\right] = \lambda. \end{aligned} \tag{11}$$

It is easy to check that strong duality holds, and the dual formulation is

$$\begin{aligned} \text{maximize:} \quad & \mathrm{Tr}\left[\rho X\right] - 1 \\ \text{subject to:} \quad & X \geq 0, \\ & \mathrm{Tr}\left[\delta X\right] \leq 1, \quad \forall \delta \in \mathcal{F}. \end{aligned} \tag{12}$$

Effectively, this means that computing the RoNM can be performed *efficiently* since we have expressed it as a SDP. However, as we'll show in Section V, the RoNM has a simple analytical form, which makes it an *optimization-free* measure.

IV. OPERATIONAL INTERPRETATION

By suitably adapting the construction of Ref. [22] for the generalized robustness of quantum channels, we provide an operational interpretation to the RoNM via a state discrimination task and a channel discrimination task. Although the original construction was meant to characterize the resourcefulness of quantum channels, this construction works for non-Markovian maps as well (which are not CP), as we will show below.

A. State discrimination

Suppose we are given an ensemble of quantum states, $\left\{p_j, \sigma_j\right\}$ and a map $\mathbb{I} \otimes \Lambda$. We are to distinguish which state σ_j has been selected from the ensemble by a single application of the map followed by a measurement with positive operator-valued measure (POVM) elements $\left\{M_j\right\}$, s.t. $M_j \geq 0, \sum_j M_j = \mathbb{I}$. The average success probability for this task is

$$p_{\text{succ}}\left(\left\{p_j, \sigma_j\right\}, \left\{M_j\right\}, \mathbb{I} \otimes \Lambda\right) = \sum_j p_j \,\mathrm{Tr}\left[\mathbb{I} \otimes \Lambda\left(\sigma_j\right) M_j\right]. \tag{13}$$

The average success probability for this task clearly depends on the ensemble, the choice of the POVMs, and the map. We show below that a maximization over all ensembles and all POVMs gives an operational characterization to the RoNM.

Theorem IV.1. *For any map Λ, Markovian or non-Markovian, it holds that*

$$\begin{aligned} \max_{\{p_j, \sigma_j\}, \{M_j\}} & \frac{p_{\text{succ}}\left(\left\{p_j, \sigma_j\right\}, \left\{M_j\right\}, \mathrm{I} \otimes \Lambda\right)}{\max\limits_{\Gamma \in \mathcal{O}_{\mathcal{F}}} p_{\text{succ}}\left(\left\{p_j, \sigma_j\right\}, \left\{M_j\right\}, \mathbb{I} \otimes \Gamma\right)} \\ & = 1 + \mathcal{N}_{\mathcal{R}}(\Lambda). \end{aligned} \tag{14}$$

Proof. Consider the optimal decomposition of Λ,

$$\Lambda = (1+s^\star)\,\Gamma - s^\star \Phi,$$

where $s^\star = \mathcal{N}_{\mathcal{R}}(\Lambda)$, and $\Gamma, \Phi \in \mathcal{O}_{\mathcal{F}}$. Then, the average success probability (see Eq. (13)) is

$$\begin{aligned} \sum_j p_j \,\mathrm{Tr}\left[\mathrm{I} \otimes \Lambda\left(\sigma_j\right) M_j\right] &= \sum_j p_j\,(1+s^\star)\,\mathrm{Tr}\left[\mathrm{I} \otimes \Gamma\left(\sigma_j\right) M_j\right] - \underbrace{\sum_j p_j s^\star \,\mathrm{Tr}\left[\mathrm{I} \otimes \Phi\left(\sigma_j\right) M_j\right]}_{\geq 0} \\ &\leq \sum_j p_j\,(1+s^\star)\,\mathrm{Tr}\left[\mathrm{I} \otimes \Gamma\left(\sigma_j\right) M_j\right] \leq (1+s^\star) \max_{\Gamma \in \mathcal{O}_{\mathcal{F}}} p_{\text{succ}}\left(\left\{p_j, \sigma_j\right\}, \left\{M_j\right\}, \mathrm{I} \otimes \Gamma\right), \end{aligned} \tag{15}$$

where $s^\star = \mathcal{N}_{\mathcal{R}}(\Lambda)$. In Eq. (15), the underbraced term is non-negative, since in our definition of the RoNM we choose $\Phi \in \mathcal{O}_{\mathcal{F}}$. However, if we were to define a *generalized* robustness measure, i.e., allow mixing with respect to any map (Markovian or non-Markovian), this would not hold true. Therefore, we have that

$$\frac{p_{\text{succ}}\left(\{p_j,\sigma_j\},\{M_j\},\mathrm{I}\otimes\Lambda\right)}{\max\limits_{\Gamma\in O_{\mathcal{F}}} p_{\text{succ}}\left(\{p_j,\sigma_j\},\{M_j\},\mathbb{I}\otimes\Gamma\right)} \leq 1+\mathcal{N}_{\mathcal{R}}(\Lambda). \tag{16}$$

Then, by maximizing over all ensembles, $\{p_j,\sigma_j\}$ and all POVMs, $\{M_j\}$, we have that the LHS of Eq. (14) is $\leq 1+\mathcal{N}_{\mathcal{R}}(\Lambda)$.

We now prove the converse part of the theorem. Recall that for a POVM element to form a valid measurement, we must have $M_j \leq \mathbb{I}\,\forall j$. Then, $\left\|M_j\right\|_\infty \leq 1$. Therefore, we consider a specific measurement

$$\left\{\frac{X}{\|X\|_\infty}, \mathbb{I}-\frac{X}{\|X\|_\infty}\right\},$$

where X is an optimal solution of the dual SDP (see Eq. (12)), and the ensemble $\{p_j,\sigma_j\}_{j=0}^1$, with $p_0=1, \sigma_0=|\Phi^+\rangle\langle\Phi^+|$, where $|\Phi^+\rangle$ is the (normalized) maximally entangled state, and $p_1=0$ and σ_1 is some arbitrary state. For this choice, we then have

$$\begin{aligned}
&\frac{p_{\text{succ}}\left(\{p_j,\sigma_j\},\{M_j\},\mathbb{I}\otimes\Lambda\right)}{\max\limits_{\Gamma_j\in O_{\mathcal{F}}} p_{\text{succ}}\left(\{p_j,\sigma_j\},\{M_j\},\mathbb{I}\otimes\Gamma\right)}\\
&= \frac{\mathrm{Tr}\left[\mathbb{I}\otimes\Lambda(|\Phi^+\rangle\langle\Phi^+|)\,X\right]}{\max\limits_{\Gamma\in O_{\mathcal{F}}}\mathrm{Tr}\left[\mathbb{I}\otimes\Gamma\left(|\Phi^+\rangle\langle\Phi^+|\right)X\right]},\\
&= \frac{\mathrm{Tr}\left[\rho_\Lambda X\right]}{\max\limits_{\Gamma\in\mathcal{O}_{\mathcal{F}}}\mathrm{Tr}\left[\rho_\Gamma X\right]} \geq \left(1+\mathcal{N}_{\mathcal{R}}(\Lambda)\right),
\end{aligned} \tag{17}$$

where the last inequality follows from the dual SDP formulation in Eq. (12).

We now have both an upper and a lower bound. From the first half of the proof, we have that the LHS of Eq. (14) is $\leq \left(1+\mathcal{N}_{\mathcal{R}}(\Lambda)\right)$. Since the LHS of Eq. (14) is (already) maximized over all measurements and ensembles, in particular, it is greater than or equal to the value for the particular ensemble and measurements chosen above (since it is the maximum). On the other hand, using the converse part of the proof (as shown by Eq. (17)), we have that the LHS of Eq. (14) is $\geq \left(1+\mathcal{N}_{\mathcal{R}}(\Lambda)\right)$. Therefore, combining these two, it must be equal to the RHS of Eq. (14), which completes the proof. ■

We see how this result gives an operational interpretation for the RoNM. For states evolving under a Markovian map, evolution always reduces the ability to distinguish the states; in general, the probability to distinguish the states decays exponentially. For non-Markovian evolution, by contrast, it is possible at times for two states to become *more* distinguishable. This increase is proportional to the robustness measure.

B. Channel discrimination

We now give another operational characterization of the robustness measure in the context of *channel discrimination*. Suppose we have access to an ensemble of maps, i.e., a quantum operation sampled from the prior distribution $\{p_j,\Lambda_j\}_{j=0}^{N-1}$. We apply one map chosen at random from this ensemble to one subsystem of a bipartite state $\Psi\in\mathcal{D}\left(\mathcal{H}\otimes\mathcal{H}\right)$, and perform a collective measurement on the output system by measurement operators $\{M_j\}_{j=0}^N$. We also allow measurements with inconclusive outcomes; for example, when using POVMs to distinguish non-orthogonal states. Then, the average success probability for this task is

$$\begin{aligned}
&\widetilde{p}_{\text{succ}}\left(\{p_j,\mathbb{I}\otimes\Lambda_j\}_{j=0}^{N-1},\{M_j\}_{j=0}^N,\Psi\right)\\
&\quad= \sum_{j=0}^{N-1} p_j\,\mathrm{Tr}\left[\{1\otimes\Lambda_j(\Psi)\}\,M_j\right],
\end{aligned} \tag{18}$$

where inconclusive measurement outcomes do not contribute to the success probability. For the ensemble of maps, we also define $\widetilde{\mathcal{N}}_{\mathcal{R}}\left(\{p_j,\Lambda_j\}\right)\equiv\max_j\mathcal{N}_{\mathcal{R}}\left(\Lambda_j\right)$. Then we can connect the maximal advantage in this channel discrimination task to the maximum robustness of the ensemble of maps.

Theorem IV.2.

$$\begin{aligned}
&\frac{\max\limits_{\Psi\in\mathcal{D}(\mathcal{H}\otimes\mathcal{H}),\{M_j\}}\widetilde{p}_{\text{succ}}\left(\{p_j,\mathbb{I}\otimes\Lambda_j\}_{j=0}^{N-1},\{M_j\}_{j=0}^N,\Psi\right)}{\max\limits_{\Gamma\in O_{\mathcal{F}}}\widetilde{p}_{\text{succ}}\left(\{p_j,\mathbb{I}\otimes\Gamma_j\}_{j=0}^{N-1},\{M_j\}_{j=0}^N,\Psi\right)}\\
&\qquad\qquad = 1+\widetilde{\mathcal{N}}_{\mathcal{R}}\left(\{p_j,\Lambda_j\}\right),
\end{aligned} \tag{19}$$

where $\mathcal{D}\left(\mathcal{H}\otimes\mathcal{H}\right)$ is the set of all bipartite density matrices.

Proof. Similar to the proof for the state discrimination game, we have,

$$\begin{aligned}
&\widetilde{p}_{\text{succ}}\left(\{p_j,\mathbb{I}\otimes\Lambda_j\}_{j=0}^{N-1},\{M_j\}_{j=0}^N,\Psi\right)\\
&= \sum_{j=0}^{N-1} p_j\,\mathrm{Tr}\left[\{\mathbb{I}\otimes\Lambda_j(\Psi)\}\,M_j\right]\\
&\leq \sum_{j=0}^{N-1} p_j\left(1+r_j\right)\mathrm{Tr}\left[\{\mathbb{I}\otimes\Gamma_j(\Psi)\}\,M_j\right]\\
&\leq \left(1+r_{j^\star}\right)\max_{\Gamma_j\in O_{\mathcal{F}}}\widetilde{p}_{\text{succ}}\left(\{p_j,\mathbb{I}\otimes\Gamma_j\}_{j=0}^{N-1},\{M_j\}_{j=0}^N,\Psi\right),
\end{aligned}$$

where $r_j := \mathcal{N}_{\mathcal{R}}(\Lambda_j)$ and $j^\star = \mathrm{argmax}_j r_j$. Once again, note that the choice of $\Phi\in\mathcal{O}_{\mathcal{F}}$ is crucial in the inequality above.

To prove the converse part of the theorem, we proceed in a similar fashion as before and make a specific choice of ensemble and POVMs. Consider, $\Psi = |\Phi^+\rangle\langle\Phi^+|$, and the POVM, $M_{j^\star} = X_{j^\star}/\left\|X_j^\star\right\|_\infty$, where $X_{j^\star}$ is an optimal witness from the dual SDP formulation for $\Lambda_{j^\star}$ (see Eq. (12)). Define, $M_j = 0$ for $j \neq j^\star$ and $M_N = \mathbb{I} - M_{j^\star}$. Then, for this specific choice of the state and measurements, we have

$$\begin{aligned}
&\frac{\widetilde{p}_{\text{succ}}\left(\{p_j, \mathbb{I}\otimes\Lambda_j\}_{j=0}^{N-1}, \{M_j\}_{j=0}^{N}, \Psi\right)}{\max\limits_{\Gamma_j\in O_{\mathcal{F}}} \widetilde{p}_{\text{succ}}\left(\{p_j, \mathbb{I}\otimes\Gamma_j\}_{j=0}^{N-1}, \{M_j\}_{j=0}^{N}, \Psi\right)}, \\
&= \frac{\sum\limits_{j=0}^{N-1} p_j \operatorname{Tr}\left[\left\{\mathbb{I}\otimes\Lambda_j\left(|\Phi^+\rangle\langle\Phi^+|\right)\right\} M_j\right]}{\max\limits_{\Gamma_j\in O_{\mathcal{F}}} \sum\limits_{j=0}^{N-1} p_j \operatorname{Tr}\left[\{\mathbb{I}\otimes\Gamma_j(|\Phi^+\rangle\langle\Phi^+|)\} M_j\right]}, \\
&= \frac{p_{j^\star}\operatorname{Tr}\left[\rho_{\Lambda_{j^\star}} X_{j^\star}\right]}{\max\limits_{\Gamma\in O_{\mathcal{F}}} p_{j^\star}\operatorname{Tr}\left[\rho_\Gamma X_{j^\star}\right]}, \\
&\leq 1 + \widetilde{\mathcal{N}}_{\mathcal{R}}(\{p_j, \Lambda_j\}),
\end{aligned} \tag{20}$$

where the last inequality is due to the dual SDP formulation (Eq. (12)). Then, by combining the converse part of the proof with the first half, the proof is complete by a similar argument as above. ■

V. RELATING THE ROBUSTNESS OF NON-MARKOVIANITY TO THE RHP MEASURE

We now relate the RoNM to the RHP measure. This relation, in turn, provides an operational meaning to the RHP measure. Consider the optimal decomposition for a Choi matrix:

$$\rho = (1+s^\star)\,\delta - s^\star\tau, \text{ where } s^\star = \mathcal{N}_{\mathcal{R}}(\rho) \text{ and } \tau, \delta \geq 0.$$

The spectral decomposition of ρ can be written

$$\rho = \Delta^+ - \Delta^-, \text{ where } \Delta^\pm \geq 0.$$

Comparing the two decompositions, we see that $\Delta^- = s^\star\tau \implies \left\|\Delta^-\right\|_1 = s^\star = \mathcal{N}_{\mathcal{R}}(\rho)$, since $\|\tau\|_1 = 1$. Also,

$$\|\rho\|_1 = \left\|\Delta^+\right\|_1 + \left\|\Delta^-\right\|_1 = \operatorname{Tr}\left[\Delta^+\right] + \operatorname{Tr}\left[\Delta^-\right]. \tag{21}$$

Since the map Λ corresponding to the Choi matrix ρ is trace-preserving, we have

$$\operatorname{Tr}[\rho] = \operatorname{Tr}\left[\Delta^+\right] - \operatorname{Tr}\left[\Delta^-\right] = 1. \tag{22}$$

Subtracting Eq. (22) from Eq. (21), we have

$$\begin{aligned}
&\|\rho\|_1 - 1 = 2\operatorname{Tr}\left[\Delta^-\right] = 2\left\|\Delta^-\right\|_1 = 2\mathcal{N}_{\mathcal{R}}(\rho) \\
&\implies \mathcal{N}_{\mathcal{R}}(\rho) = \frac{\|\rho\|_1 - 1}{2} = \frac{f_{\text{NCP}} - 1}{2},
\end{aligned} \tag{23}$$

where

$$f_{\text{NCP}} := \left\|\left(\mathcal{E}_{(t+\epsilon,t)}\otimes\mathbb{I}\right)(|\Phi\rangle\langle\Phi|)\right\|_1$$

in the RHP construction (see page 3 of Ref. [7]).

Note that, by construction, ρ is a function of both t and ϵ. Therefore, to compare to the non-Markovianity measure of RHP, we'll need to remove the ϵ dependence and integrate over time, where the only contribution comes from the times at which the map (and hence the measure) is non-Markovian. That is, define

$$\mathcal{N}_{\mathcal{R}}^{\text{total}} = \int_0^\infty \left(\lim_{\epsilon\to 0^+} \frac{\mathcal{N}_{\mathcal{R}}(\rho)}{\epsilon}\right) dt.$$

Comparing to the RHP measure, we have

$$\mathcal{N}_{\mathcal{R}}^{\text{total}} = \frac{\mathcal{I}}{2}, \tag{24}$$

where $\mathcal{I}$ is the RHP measure for non-Markovianity (see page 3 of Ref. [7]). Moreover, one can also define a normalized version of the measure here (in analogy to the normalized measure of RHP). Define $\mathcal{N}_{\mathcal{R}}^{\text{norm}} = \frac{\mathcal{N}_{\mathcal{R}}^{\text{total}}}{1+\mathcal{N}_{\mathcal{R}}^{\text{total}}}$. Therefore, $\mathcal{N}_{\mathcal{R}}^{\text{norm}} = 0$ for $\mathcal{N}_{\mathcal{R}}^{\text{total}} = 0$ (Markovian evolution) and $\mathcal{N}_{\mathcal{R}}^{\text{norm}} \to 1$ for $\mathcal{N}_{\mathcal{R}}^{\text{total}} \to \infty$. Note that the RoNM is purely operationally-motivated while the RHP is (purely) *physically* motivated. The fact that the two measures are exactly equal (up to a factor of one half) is quite remarkable.

VI. ANALYTICAL FORM FOR SINGLE QUBIT-CHANNELS

We now consider a single-qubit dephasing channel as an analytical example. The Lindblad equation for this channel has the form:

$$\frac{d\rho}{dt} = \gamma(t)\left(\sigma_z\rho\sigma_z - \rho\right), \tag{25}$$

where $\gamma(t)$ is the rate of dephasing. In the small time approximation, the eigenvalues of the Choi matrix are $\{0, 0, \epsilon\gamma(t), 1-\epsilon\gamma(t)\}$, respectively. For Markovian operations, $\gamma(t) \geq 0$, but for non-Markovian operations $\gamma(t)$ can be negative. Note that for small ϵ, if the rate is negative then all the eigenvalues of ρ are positive except for $\epsilon\gamma(t)$. Then, we have,

$$g(t) = \begin{cases} 0 & \gamma(t) \geq 0, \\ -2\gamma(t) & \gamma(t) < 0, \end{cases} \tag{26}$$

and

$$\lim_{\epsilon\to 0^+} \frac{\mathcal{N}_{\mathcal{R}}(\rho)}{\epsilon} = \begin{cases} 0 & \gamma(t) \geq 0, \\ -\gamma(t) & \gamma(t) < 0. \end{cases}. \tag{27}$$

As a result, $\mathcal{N}_{\mathcal{R}}^{\text{total}} = \mathcal{I}/2$, for the dephasing channel, where $\mathcal{I}$ is the RHP measure. We list some more examples in Appendix C.

VII. DISCUSSION

In this work, we have constructed a resource-theoretic measure of quantum non-Markovianity (in the CP-divisibility sense) with a direct operational interpretation: the Robustness of Non-Markovianity (RoNM). By identifying a meaningful set of *free operations* and carefully characterizing quantum non-Markovianity, we constructed a measure that is faithful, convex, and monotonic. Using the semidefinite programming form for the RoNM, we established an operational interpretation by both a state and a channel discrimination task. Moreover, we obtained an optimization-free, closed-form expression for this measure.

Remarkably, the RoNM measure (which is motivated by its operational interpretation) turns out to be exactly one half of the RHP measure, which is physically motivated by studying the system-ancilla entanglement dynamics [7]. This intriguing connection was obtained by using the powerful results that underlie quantum resource theories, which speaks volumes about the efficacy of the resource-theoretic approach to characterizing quantum resources. Furthermore, this connection provides a direct operational meaning to the well-known RHP measure in terms of the channel and state discrimination tasks above. Moreover, not only does the RHP measure borrow the operational meaning of the RoNM, but the RoNM inherits the physical interpretation of the RHP measure.

Several open questions emerge from our work: for example, one can construct Schatten p-norm based quantifiers of non-Markovianity; which measure the distance of a non-Markovian map from the set of free (Markovian) channels. Using the gauge functions formalism [32], one can relate these (and other) measures to the RoNM and it will be interesting to see if the operational interpretation is carried over via these relations. Second, are there other operational measures—motivated from quantum resource theories—that can quantify non-Markovianity and if yes, how do they relate to the RoNM? Moreover, how does the RoNM connect to the different quantifiers of non-Markovianity, like those based on the quantum fisher information [33], degree of non-Markovianity [34], relative entropy of coherence [35], quantum interferometric power [36], etc. Finally, future work(s) will also explore the problem of characterizing the information backflow approach to non-Markovianity [6] using our resource-theoretic construction.

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Appendix A: Choi-Jamiołkowski isomorphism

Using Theorem 3.3 and 3.4 of Ref. [3], we have

$$\mathcal{E} \text{ is CP} \iff \left\| [\mathcal{E} \otimes \mathbb{I}] \left(\tilde{\Delta}\right) \right\|_1 \leqslant \|\tilde{\Delta}\|_1, \tag{A1}$$

for $\widetilde{\Delta} \in \text{Herm}(\mathcal{H} \otimes \mathcal{H})$. However, if $\mathcal{E}$ is not CP, then, we will have

$$\left\| [\mathcal{E} \otimes \mathbb{I}] \left(\tilde{\Delta}\right) \right\|_1 > \|\tilde{\Delta}\|_1. \tag{A2}$$

Since $|\Phi\rangle\langle\Phi|$ is positive semidefinite, we have,

$$\left\| [\mathcal{E} \otimes \mathbb{I}] \left(|\Phi\rangle\langle\Phi|\right) \right\|_1 \begin{cases} = 1 & \text{iff } \mathcal{E} \text{ is CP} \\ > & 1 \text{ otherwise.} \end{cases} \tag{A3}$$

Appendix B: Proof that the set of Markovian Choi matrices are a convex and compact set in the small-time limit

Theorem B.1 (Ref. [26]). *The set of all Markovian Choi matrices,*

$$\mathcal{F}^{\epsilon,t} = \left\{ \rho_\Phi(t+\epsilon, t) \mid \left\| \rho_\Phi(t+\epsilon, t) \right\|_1 = 1, \forall t, \epsilon > 0 \right\}$$

is a convex and compact set in the limit $\epsilon \to 0^+$.

Proof. Consider two Markovian channels,

$$\Lambda^{(i)}(t+\epsilon, t) \equiv \mathcal{T} \exp\left(\int_t^{t+\epsilon} \mathcal{L}_\tau^{(i)} d\tau \right), \; i = 1, 2. \tag{B1}$$

For sufficiently small ϵ, we can Taylor expand the exponential. Then, neglecting terms of $O(\epsilon^2)$ and higher, we have

$$\Lambda^{(i)}(t+\epsilon, t) = \mathbb{I} + \epsilon \mathcal{L}_t^{(i)}, \; i = 1, 2. \tag{B2}$$

A convex combination of these maps is

$$\begin{aligned} \Lambda(t+\epsilon, t) &= p\Lambda^{(1)}(t+\epsilon, t) + (1-p)\Lambda^{(2)}(t+\epsilon, t) \\ &= \mathbb{I} + \epsilon[p\mathcal{L}_t^{(1)} + (1-p)\mathcal{L}_t^{(2)}] = \mathbb{I} + \epsilon\mathcal{L}_t, \end{aligned} \tag{B3}$$

with $\mathcal{L}_t = p\mathcal{L}_t^{(1)} + (1-p)\mathcal{L}_t^{(2)}$ and $0 \leq p \leq 1$. That is, we have a Lindblad type generator $\mathcal{L}_t$ with positive coefficients. Therefore, $\Lambda(t+\epsilon, t) \in \mathcal{O}_\mathcal{F}$, which implies that the set of Markovian channels forms a convex set. Using the Choi-Jamiołkowski isomorphism, the Choi matrices corresponding to the Markovian channels also form a convex set.

As for compactness, it is easy to prove that $\mathcal{F}^{\epsilon,t}$ is closed and bounded using the continuity of the 1-norm, along with the fact that all norms are equivalent in finite dimensions. ■

Appendix C: Some more analytical examples

1. Pauli channel

Consider the Pauli channel,

$$\frac{d\rho}{dt} = \gamma_x(t)\left(\sigma_x\rho\sigma_x - \rho\right) + \gamma_y(t)\left(\sigma_y\rho\sigma_y - \rho\right) + \tag{C1}$$

$$\gamma_z(t)\left(\sigma_z\rho\sigma_z - \rho\right), \tag{C2}$$

where $\gamma_i(t)$, $i = x, y, z$, are the rates of the different dephasing processes. Then we have

$$\lim_{\epsilon\to 0^+} \frac{\mathcal{N}_\mathcal{R}(\rho)}{\epsilon} = \begin{cases} 0 & \text{for } \gamma_i(t) \geq 0, \; i = x, y, z \\ -\sum_i \gamma_i(t) & \text{for } \gamma_i(t) < 0, \; i = x, y, z \end{cases}. \tag{C3}$$

2. Abrupt transitions between Markovianity and non-Markovianity

Consider the single qubit model studied by Pang *et. al* in Ref. [37]. The usual understanding of Markovian dynamics is that it represents an asymptotic limiting behavior under the Born-Markov approximation. However, in Ref. [37] the authors have obtained some surprising results that call that usual understanding into doubt. In particular, by continuously refreshing the bath qubit state, they show that there is a sharp threshold for the cooling rate at which the system transitions from non-Markovian to Markovian dynamics.

We rewrite their single-qubit master equation in Lindblad form:

$$\partial_t \rho_t^{\text{S}} = -\frac{\dot{c}_t}{2c_t}\left(\sigma_x^{\text{S}}\rho_t^{\text{S}}\sigma_x^{\text{S}} - \rho_t^{\text{S}}\right). \tag{C4}$$

Since the master equation is identical to a dephasing channel in the σ_x basis, we see that the non-Markovian nature of the evolution can be readily captured by our measure. In their example, the dephasing rates become negative when $\kappa^2 < 64\xi^2$, which is precisely when our non-Markovianity measure would become nonzero. This is consistent with their analysis of the distinguishability measure [6], which is non-negative in the same parameter regime. Therefore, our RoNM measure captures the non-Markovianity of this nontrivial example as well.